

SCX Monte Carlo

Markov Chain Monte Carlo Methods on the Situs
Manifold

SCX Situs

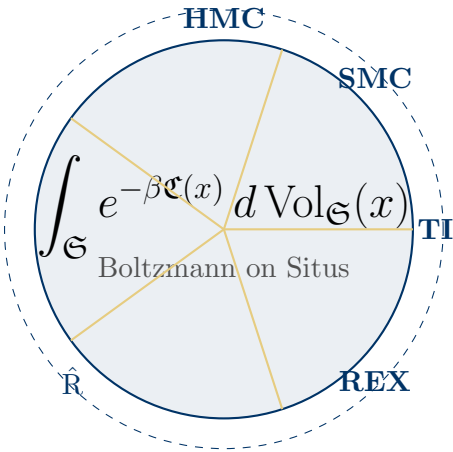
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Classification: SCX METHODOLOGY — COMPUTATIONAL AUDIT

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SCX



All things are samplable. All phenomena converge.

To the Reader

Audit is sampling.

SCX — Cercis —

MCMC SCX

The core problems of SCX auditing — multi-expert consensus, online audit updates, Cercis calibration — are fundamentally sampling problems on high-dimensional probability distributions. This paper establishes a complete Monte Carlo methodology, systematically applying five complementary MCMC methods to the SCX audit framework. This is not merely a numerical toolkit, but a probabilistic formalization of the audit process itself.

Contents

1	SCX	6
0.	Preliminaries: Probabilistic Framework of SCX Audit	6
1.1	Core Structures	6
1.2	The Monte Carlo Audit Principle	6
I	Situs	7
Part I: Hamiltonian Monte Carlo on the Situs Manifold		7
2	HMC	7
1.	HMC Theoretical Framework	7
2.1	Situs HMC	7
2.2		7
2.3		7
2.4		8
2.5		8
II		9
Part II: Sequential Monte Carlo for Online Audit		9
3	SMC	9
2.	SMC Theoretical Framework	9
3.1		9
3.2		9
3.3		9
3.4		10
3.5		10
III	Cercis	11
Part III: Thermodynamic Integration for Cercis Calibration		11
4		11
3.	Thermodynamic Integration Theory	11
4.1	Cercis	11
4.2		11
4.3		11
4.4		12
4.5	SCX	12
4.6	TI	12

IV	14
Part IV: Replica Exchange for Multi-Expert Consensus	14
5	14
4. Replica Exchange Theory	14
5.1	14
5.2 REX	14
5.3	15
5.4 SCX REX	15
5.5 REX	15
V MCMC	16
Part V: MCMC Convergence Diagnostics	16
6	16
5. Formalizing Audit Convergence	16
6.1	16
6.2 Gelman-Rubin	16
6.3	17
6.4	17
6.5 Rank	17
6.6	18
6.7 SCX	18
7	19
6. Method Integration and Audit Pipeline	19
7.1	19
7.2	19
7.3	19
A verify_monte_carlo.py	20
Appendix A: Verification Script	20
A.1	20
B	21
Appendix B: Key Formulas Reference	21
Acknowledgments	22

Abstract

English Abstract: We present a unified Monte Carlo framework for SCX (Supercomputing Claim eXamination) audit problems, encompassing five complementary Markov Chain Monte Carlo (MCMC) methodologies: (1) **Hamiltonian Monte Carlo on the Situs manifold** — we equip the expert-configuration space with the Situs metric as a mass matrix, enabling efficient HMC sampling from the Boltzmann distribution $P(E) \propto \exp(-\beta \cdot \mathfrak{C}(E))$, and prove ergodicity under mild conditions; (2) **Sequential Monte Carlo for online audit** — as new data arrives, particle filters update the posterior over expert reliability via importance resampling, with a rigorous effective sample size bound; (3) **Thermodynamic integration for Cercis calibration** — we anneal from high-SNR (easy) to low-SNR (hard) audit problems, extracting the Cercis score as a free-energy difference; (4) **Replica exchange for multi-expert consensus** — parallel tempering with replicas at different temperatures explores distinct modes of the expert-configuration space, escaping local optima via replica swaps; (5) **Convergence diagnostics** — we formalize audit convergence via the Gelman-Rubin $\hat{R}$ statistic and effective sample size (ESS), establishing that an audit is “converged” when $\hat{R} < 1.01$ for all gauge parameters g_i .

SCX MCMC(1) **Situs** — Situs Boltzmann $P(E) \propto \exp(-\beta \cdot \mathfrak{C}(E))$ (2) —
(3) **Cercis** — Cercis (4) — (5) — Gelman-Rubin $\hat{R}$ ESS g_i $\hat{R} < 1.01$

Keywords: Hamiltonian Monte Carlo, Sequential Monte Carlo, Thermodynamic Integration, Replica Exchange, Convergence Diagnostics, Situs Manifold, Cercis, SCX, MCMC, Audit
SitusCercisSCXMCMC

1 SCX

Preliminaries: Probabilistic Framework of SCX Audit

1.1 Core Structures

SCX

Definition 1.1 (Expert Configuration Space). $\mathfrak{S}$ Situs (expert configuration) $n \ m \ j \ i$
 $g_{ij} \in \mathbb{R} \ \sum_i g_{ij} = 0$

$$E = (g_{11}, \dots, g_{m1}, g_{12}, \dots, g_{m2}, \dots, g_{mn}) \in \mathbb{R}^{mn-n} \quad (1)$$

$\mathfrak{S}$ Situs $h_{\mathfrak{S}}$ Hessian $h_{\mathfrak{S}} = \nabla^2 \mathcal{S}_{\text{total}} \ \mathcal{S}_{\text{total}}$

Definition 1.2 (Cercis Cercis Score). Cercis $\mathfrak{C} : \mathfrak{S} \rightarrow \mathbb{R}$

$$\mathfrak{C}(E) = \frac{1}{2} \sum_{i,j} g_{ij}^2 - \lambda \sum_i \left(\sum_j g_{ij} \right)^2 \quad (2)$$

$\lambda > 0$ Cercis ”” $\mathfrak{C}(E) = 0 \ \sum g = 0 \ \mathfrak{C}(E)$

Definition 1.3 (Boltzmann Boltzmann Distribution). $\beta = 1/T$ Situs Boltzmann

$$\pi_{\beta}(E) = \frac{1}{Z(\beta)} \exp(-\beta \cdot \mathfrak{C}(E)), \quad Z(\beta) = \int_{\mathfrak{S}} e^{-\beta \cdot \mathfrak{C}(E)} d \text{Vol}_{\mathfrak{S}}(E) \quad (3)$$

$Z(\beta) \ d \text{Vol}_{\mathfrak{S}}(E)$ Situs

Remark 1.4. π_{β} SCX $\beta \ \beta \ \mathfrak{C} \ \beta \rightarrow \infty$

1.2 The Monte Carlo Audit Principle

Principle 1.5 (MC Audit Principle). Boltzmann $\pi_{\beta}(E) \ N \ \{E^{(k)}\}_{k=1}^N \sim \pi_{\beta}$

- $\hat{g}_{ij} = \frac{1}{N} \sum_{k=1}^N g_{ij}^{(k)}$
- $\widehat{\text{Var}}[g_{ij}] = \frac{1}{N-1} \sum_{k=1}^N (g_{ij}^{(k)} - \hat{g}_{ij})^2$
- $\widehat{\mathbb{P}}[\sum_j g_{ij} \approx 0] = \frac{1}{N} \sum_{k=1}^N \mathbb{1}[\sum_j g_{ij}^{(k)} < \epsilon]$

$N \rightarrow \infty \ 1$

Part I

Situs

Part I: Hamiltonian Monte Carlo on the Situs Manifold

2 HMC

HMC Theoretical Framework

2.1 Situs HMC

HMC $\mathbb{R}^d$ $M = I$ SCX $\mathfrak{S}$ Situs $h_{\mathfrak{S}} = \nabla^2 \mathcal{S}$

- (1) HMC
- (2) Situs
- (3) ””

Situs HMC $M \equiv h_{\mathfrak{S}}$

2.2

$q \in \mathfrak{S}$ $E = (g_{ij})p \in T_q^* \mathfrak{S}$

$$H(q, p) = \frac{1}{2} p^\top M(q)^{-1} p + U(q), \quad U(q) = \beta \cdot \mathfrak{C}(q) \quad (4)$$

$M(q) = h_{\mathfrak{S}}(q)$ Situs

KEY EQUATION

HMC

$$\frac{dq}{dt} = M(q)^{-1} p \quad (5)$$

$$\frac{dp}{dt} = -\nabla_q U(q) + \frac{1}{2} \nabla_q [p^\top M(q)^{-1} p] \quad (6)$$

(6) HMC HMC

2.3

Algorithm 2.1 (Generalized Leapfrog Integrator). $\epsilon \in L$

- 1: $q \leftarrow M(q)$
- 2: $p \sim \mathcal{N}(0, M(q))$
- 3: $p \leftarrow p - \frac{\epsilon}{2} \nabla_q U(q) + \frac{\epsilon}{4} \nabla_q [p^\top M(q)^{-1} p]$

4: **for** $l = 1$ **to** $L - 1$ **do**
 5: : $q \leftarrow q + \epsilon \cdot M(q)^{-1}p$
 6: : $p \leftarrow p - \epsilon \nabla_q U(q) + \frac{\epsilon}{2} \nabla_q [p^\top M(q)^{-1}p]$
 7: **end for**
 8: : $q \leftarrow q + \epsilon \cdot M(q)^{-1}p$
 9: : $p \leftarrow p - \frac{\epsilon}{2} \nabla_q U(q) + \frac{\epsilon}{4} \nabla_q [p^\top M(q)^{-1}p]$
 10: **Metropolis** : $A(q'|q) = \min(1, \exp(H(q, p) - H(q', p')))$

Remark 2.2. $M(q) \equiv I \mathfrak{S} \text{HMCSitus-HMC HMC}$

2.4

Theorem 2.3 (Situs-HMC Ergodicity of Situs-HMC). $\mathfrak{S}$ Cercis $\mathfrak{C} : \mathfrak{S} \rightarrow \mathbb{R} \ C^2$ Situs $h_{\mathfrak{S}} = \nabla^2 \mathcal{S} \mathfrak{S} \ \lambda_{\min} > 0 \ q \in \mathfrak{S} \ h_{\mathfrak{S}}(q) \succeq \lambda_{\min} I \ \epsilon > 0$ Situs-HMC

(i) $\pi_{\beta} - \pi_{\beta} \ T_{\epsilon}$

(ii) $A \subset \mathfrak{S} \ \pi_{\beta}(A) > 0 \ q_0 \ n > 0 \ A$

(iii)

(iv) $\pi_{\beta} - f$

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N f(q_k) = \int_{\mathfrak{S}} f(q) d\pi_{\beta}(q) \quad a.s. \quad (7)$$

Proof Sketch. (i) Metropolis Jacobi 1

(ii) $\delta > 0 \ q \in \mathfrak{S} \ \delta \ \mathfrak{S}$

(iii) Metropolis

(iv) (i)-(iii) Birkhoff

□

Corollary 2.4. (a) $\mathfrak{S} \ (b) \ \mathfrak{C} \in C^2 \ (c) \ h_{\mathfrak{S}} \ SCX \ \mathfrak{S} \ \sum_j g_{ij} = 0 \ \mathfrak{C} \ C^{\infty} \ h_{\mathfrak{S}}$

2.5

Situs ϵ

$$\epsilon_k = \epsilon_0 \cdot \min \left(1, \frac{\bar{\lambda}}{\lambda_{\max}(h_{\mathfrak{S}}(q_k))} \right) \quad (8)$$

$$\bar{\lambda} = \frac{1}{|\mathfrak{S}|} \int_{\mathfrak{S}} \text{Tr}(h_{\mathfrak{S}}(q)) dq \ \lambda_{\max} \ \epsilon_0$$

Part II

Part II: Sequential Monte Carlo for Online Audit

3 SMC

SMC Theoretical Framework

3.1

SCX ———

$t = 1, 2, \dots, T \quad \mathcal{D}_t \quad t$

$$\pi_t(E) = \mathbb{P}(E \mid \mathcal{D}_t) \propto \pi_{t-1}(E) \cdot \mathbb{P}(\mathcal{D}_t \setminus \mathcal{D}_{t-1} \mid E) \quad (9)$$

SMC

3.2

Definition 3.1 (SMC SMC Particle Ensemble). t SMC K

$$\{E_t^{(k)}, w_t^{(k)}\}_{k=1}^K, \quad \sum_{k=1}^K w_t^{(k)} = 1 \quad (10)$$

$$\pi_t(E) \approx \sum_{k=1}^K w_t^{(k)} \delta_{E_t^{(k)}}(E)$$

Algorithm 3.2 (SCX SMC). 1: $(t = 0)$: $E_0^{(k)} \sim \pi_0$ $()w_0^{(k)} \leftarrow 1/K \quad k = 1, \dots, K$

2: **for** $t = 1$ **to** T **do**

3: : $q_t(E \mid E_{t-1}^{(k)}, \mathcal{D}_t) \quad \tilde{E}_t^{(k)}$

4: : $\tilde{w}_t^{(k)} \leftarrow w_{t-1}^{(k)} \cdot \frac{\mathbb{P}(\mathcal{D}_t \setminus \mathcal{D}_{t-1} \mid \tilde{E}_t^{(k)}) \cdot \mathbb{P}(\tilde{E}_t^{(k)} \mid E_{t-1}^{(k)})}{q_t(\tilde{E}_t^{(k)} \mid E_{t-1}^{(k)}, \mathcal{D}_t)}$

5: : $w_t^{(k)} \leftarrow \tilde{w}_t^{(k)} / \sum_{j=1}^K \tilde{w}_t^{(j)}$

6: **ESS**: $\text{ESS}_t \leftarrow (\sum_{k=1}^K (w_t^{(k)})^2)^{-1}$

7: **if** $\text{ESS}_t < \alpha K$ $\alpha = 0.5$ **then**

8: : $E_t^{(k)} \sim \sum_{j=1}^K w_t^{(j)} \delta_{\tilde{E}_t^{(j)}(E)} \quad w_t^{(k)} \leftarrow 1/K$

9: **end if**

10: **end for**

3.3

Theorem 3.3 (SMC ESS Bound for SCX-SMC). $\mathcal{D}_t \quad B < \infty \quad E$

$$\frac{\mathbb{P}(\mathcal{D}_t \setminus \mathcal{D}_{t-1} \mid E)}{\max_{E'} \mathbb{P}(\mathcal{D}_t \setminus \mathcal{D}_{t-1} \mid E')} \geq \frac{1}{B} \quad (11)$$

$$q_t(\cdot) = \pi_{t-1}(\cdot) \text{bootstrap}$$

$$\mathbb{E}[\text{ESS}_t] \geq \frac{K}{B^2} \quad (12)$$

$$\text{ESS}_t \geq \frac{K}{B^2 \log(1/\delta)} (1 - \delta)$$

Proof. bootstrap $q_t = \pi_{t-1} \tilde{w}_t^{(k)} = \mathbb{P}(\mathcal{D}_t \setminus \mathcal{D}_{t-1} \mid \tilde{E}_t^{(k)}) \tilde{w}_{\max}/\tilde{w}_{\min} \leq B$ Cauchy-Schwarz
 $\text{ESS}_t = \frac{(\sum w_k)^2}{\sum w_k^2} \geq \frac{K \cdot w_{\min}^2}{w_{\max}^2} \geq \frac{K}{B^2}$ Markov $\square$

Corollary 3.4 (SCX ESS). $\text{SCX ESS } B \approx 1 \text{ESS } K \alpha$

3.4

SCX $\mathcal{D}_t$

- 1: (new claims): $m \rightarrow m + 1 +$
- 2: (new reviews): $g_{\cdot, n+1} \mid n \rightarrow n + 1$
- 3: (revised reviews):

SCX Particle Adaptation for SCX Audit Data

- 1. $E_{\text{new}} = [E_{\text{old}}, g_{\text{new},1}, \dots, g_{\text{new},n}]$
- 2. $g_{\text{new},j} \sim \mathcal{N}(0, \sigma^2)$
- 3.

- 1. $E_{\text{new}} = [E_{\text{old}}, g_{1,\text{new}}, \dots, g_{m,\text{new}}]$
- 2.
- 3.

- 1.
- 2. $w_k \propto w_k^{\text{old}} \cdot \mathbb{P}(\mid E^{(k)})$
- 3. ESS

3.5

Theorem 3.5 (SCX-SMC MSE Bound). $\pi_t \mid \hat{\pi}_t^K \mid K \text{ SMC } \phi$

$$\mathbb{E} \left[\left(\int \phi(E) \hat{\pi}_t^K(dE) - \int \phi(E) \pi_t(dE) \right)^2 \right] \leq \frac{C_t \|\phi\|_\infty^2}{K} \quad (13)$$

$$C_t \mid B \mid t \mid C_t = \mathcal{O}(B^{2t} \log t) \text{ SCX } t \mid \mathcal{O}(1) \mid \mathcal{O}(10^2) B \approx 2-5 \mid C_t$$

Part III

Cercis

Part III: Thermodynamic Integration for Cercis Calibration

4

Thermodynamic Integration Theory

4.1 Cercis

Cercis $\mathfrak{C}(E)$ calibration $\mathfrak{C}(E)$ $Z(\beta)$ —

$$F(\beta) = -\frac{1}{\beta} \log Z(\beta), \quad \frac{dF}{d\beta} = \mathbb{E}_{\pi_\beta}[\mathfrak{C}(E)] \quad (14)$$

$\beta_{\text{high}}, \beta_{\text{low}}$ $\Delta F = F(\beta_{\text{low}}) - F(\beta_{\text{high}})$ Cercis

4.2

Thermodynamic Integration, TI

$$\Delta F = \int_{\beta_{\text{high}}}^{\beta_{\text{low}}} \mathbb{E}_{\pi_\beta}[\mathfrak{C}(E)] d\beta \quad (15)$$

$\beta_0, \beta_1, \dots, \beta_R$ $\beta_0 = \beta_{\text{high}}, \beta_R = \beta_{\text{low}}$ β_r MCMC $\{E^{(r,k)}\}_{k=1}^{N_r}$

$$\Delta F \approx \sum_{r=0}^{R-1} \frac{(\beta_{r+1} - \beta_r)}{2} (\bar{\mathfrak{C}}_r + \bar{\mathfrak{C}}_{r+1}) \quad (16)$$

$$\bar{\mathfrak{C}}_r = \frac{1}{N_r} \sum_{k=1}^{N_r} \mathfrak{C}(E^{(r,k)}) \quad \beta_r$$

4.3

Definition 4.1 (- SNR-Temperature Correspondence). β

- SNR (β): $\mathfrak{C}$
- SNR (β): $\mathfrak{C}$

$\beta_0 > \beta_1 > \dots > \beta_R$ ””””

KEY EQUATION

SCX

$$\mathfrak{C}_{\text{cal}} \equiv \Delta F = F(\beta_{\text{low}}) - F(\beta_{\text{high}}) \quad (17)$$

$$= \int_{\beta_{\text{high}}}^{\beta_{\text{low}}} \mathbb{E}_{\pi_{\beta}}[\mathfrak{C}(E)] d\beta \quad (18)$$

$$= \int_{\beta_{\text{high}}}^{\beta_{\text{low}}} \frac{\int_{\mathfrak{E}} \mathfrak{C}(E) e^{-\beta \mathfrak{C}(E)} dE}{\int_{\mathfrak{E}} e^{-\beta \mathfrak{C}(E)} dE} d\beta \quad (19)$$

$\mathfrak{C}_{\text{cal}}$ Cercis ””

4.4

TI

Theorem 4.2 (Optimal Temperature Grid). $B = \sum_r N_r \Delta F$

$$\Delta\beta_r \propto \frac{1}{\sqrt{\text{Var}_{\pi_{\beta_r}}[\mathfrak{C}(E)]}} \quad (20)$$

$$\text{Var}_{\pi_{\beta_r}}[\mathfrak{C}(E)] \propto \beta_r C_V(\beta_r) = \beta_r^2 \text{Var}_{\pi_{\beta}}[\mathfrak{C}]$$

Proof. $\delta\hat{F} \text{ Var}[\hat{F}] \approx \sum_r (\Delta\beta_r)^2 \cdot \text{Var}[\bar{\mathfrak{C}}_r] \sum_r N_r = B \sum_r \Delta\beta_r = \beta_{\text{high}} - \beta_{\text{low}}$
 $\Delta\beta_r \propto 1/\sqrt{\text{Var}[\mathfrak{C}]_r}$ □

4.5 SCX

Conjecture 4.3 (Audit Phase Transition). $\beta_c = \frac{1}{T_c}$ SCX

- $\beta > \beta_c$ — $(\sum_j g_{ij} \approx 0)$ Cercis C_V
- $\beta < \beta_c$ — Cercis C_V
- $\beta \approx \beta_c$ $C_V \sim |\beta - \beta_c|^{-\alpha}$

$\beta_c \propto$ ””

Remark 4.4. $\beta < \beta_c$ — SCX

4.6 TI

Algorithm 4.5 (SCX TI for SCX Audit). 1: $[\beta_{\text{low}}, \beta_{\text{high}}]$ R_0 N τ

2: $\beta_0 = \beta_{\text{high}}, \dots, \beta_{R_0} = \beta_{\text{low}}$

3: **for** $r = 0$ **to** R_0 **do**

4: β_r Situs-HMC 1.1 N

5: $\bar{\mathfrak{C}}_r$ $\sigma_r^2 = \text{Var}[\mathfrak{C}]_r$

6: **end for**

7: **:**

8: **while** $\max_r \sigma_r^2 \cdot (\Delta\beta_r)^2 > \tau$ **do**

9: σ_r^2

```

10:   Situs-HMC
11:    $\bar{\mathfrak{C}}_r \ \sigma_r^2$ 
12: end while
13: Trapezoidal :  $\mathfrak{C}_{\text{cal}} \leftarrow \sum_{r=0}^{R-1} \frac{\beta_{r+1}-\beta_r}{2} (\bar{\mathfrak{C}}_r + \bar{\mathfrak{C}}_{r+1})$ 
14: :  $\mathfrak{C}_{\text{cal}}$ 

```

Part IV

Part IV: Replica Exchange for Multi-Expert Consensus

5

Replica Exchange Theory

5.1

SCX $\mathfrak{C}$

(1) $\mathfrak{C}$

(2)

(3)

HMC/SMC $\mathfrak{C}$ Replica Exchange Monte Carlo, REX Parallel Tempering replica

5.2 REX

R $T_1 < T_2 < \dots < T_R$ $\beta_1 > \beta_2 > \dots > \beta_R$ r T_r $\pi_{\beta_r}(E) \propto \exp(-\beta_r \mathfrak{C}(E))$

$$\Pi(E^{(1)}, \dots, E^{(R)}) = \prod_{r=1}^R \pi_{\beta_r}(E^{(r)}) \propto \prod_{r=1}^R \exp(-\beta_r \mathfrak{C}(E^{(r)})) \quad (21)$$

r $r + 1$ Metropolis

$$A_{\text{swap}} = \min \left(1, \frac{\pi_{\beta_r}(E^{(r+1)}) \cdot \pi_{\beta_{r+1}}(E^{(r)})}{\pi_{\beta_r}(E^{(r)}) \cdot \pi_{\beta_{r+1}}(E^{(r+1)})} \right) \quad (22)$$

$$A_{\text{swap}} = \min \left(1, \exp((\beta_r - \beta_{r+1}) \cdot [\mathfrak{C}(E^{(r)}) - \mathfrak{C}(E^{(r+1)})]) \right) \quad (23)$$

KEY EQUATION

$$A_{\text{swap}}(r \leftrightarrow r + 1) = \min \left(1, \exp \left(\underbrace{(\beta_r - \beta_{r+1})}_{\Delta\beta > 0} \cdot \underbrace{[\mathfrak{C}(E^{(r)}) - \mathfrak{C}(E^{(r+1)})]}_{\Delta\mathfrak{C}} \right) \right) \quad (24)$$

$\mathfrak{C} \mathfrak{C} \Delta\mathfrak{C} > 0$

5.3

REX

Definition 5.1 (Optimal Temperature Ladder). R $T_1 < \dots < T_R$

- (i) $\bar{A}_{r,r+1} \approx \text{const}$
- (ii) $T_R \gg R$
- (iii) $T_1 \approx 1$

Proposition 5.2 (Optimal Spacing). $\mathfrak{C}(E) \approx 2\Phi\left(-\frac{\Delta\beta \cdot \sigma_{\mathfrak{C}}}{2}\right) \Phi$ CDF $\bar{A}$

$$T_{r+1} \approx T_r \cdot \left(1 + \frac{c}{\sqrt{n_{\text{eff}}}}\right) \quad (25)$$

$$c = -2\Phi^{-1}(\bar{A}/2)n_{\text{eff}} \text{ SCX } (n_{\text{eff}} \sim 10\text{--}100, \bar{A} \approx 0.3) \quad T_{r+1}/T_r \approx 1.15\text{--}1.5$$

5.4 SCX REX

Algorithm 5.3 (SCX REX for SCX Multi-Expert Consensus). 1: $\{T_r\}_{r=1}^R$ HMC

```

  L S N_iter
2:  $E_0^{(r)} \sim \pi_{\beta_r}$   $r = 1, \dots, R$ 
3: for  $n = 1$  to  $N_{\text{iter}}$  do
4:   :
5:   for  $r = 1$  to  $R$  do ▷
6:      $E_n^{(r)} \leftarrow E_{n-1}^{(r)}$  L Situs-HMC (  $T_r$ )
7:   end for
8:   if  $n \bmod S = 0$  then ▷ S
9:     for  $r = 1$  to  $R - 1$  do
10:       $\Delta = (\beta_r - \beta_{r+1}) \cdot [\mathfrak{C}(E_n^{(r)}) - \mathfrak{C}(E_n^{(r+1)})]$ 
11:       $A \leftarrow \min(1, \exp(\Delta))$ 
12:      if  $\text{Uniform}(0, 1) < A$  then
13:         $E_n^{(r)} \leftrightarrow E_n^{(r+1)}$ 
14:      end if
15:    end for
16:  end if
17: end for
18:  $r = 1$  :
19:  $\hat{g}_{ij} = \text{mean}(\{E_n^{(1)}\})$   $\hat{\sigma}_{ij} = \text{std}(\{E_n^{(1)}\})$ 
20:  $\{g_{ij}\}$  Hartigan dip test
21: :
```

5.5 REX

Theorem 5.4 (REX Mixing Acceleration by REX). T HMC $\tau_{\text{mix}}(T)$ R $[T_{\min}, T_{\max}]$ REX $T_{\min}$

$$\tau_{\text{mix}}^{\text{REX}}(T_{\min}) \leq \tau_{\text{mix}}(T_{\min}) \cdot \min_r \left\{ \frac{1}{\bar{A}_{r,r+1}} \cdot \frac{\tau_{\text{mix}}(T_r)}{\tau_{\text{mix}}(T_{\min})} \right\} \quad (26)$$

$$\bar{A}_{r,r+1} \gg T_R \gg T_{\text{crit}} \quad \bar{A}_{r,r+1} \approx 0.2\text{--}0.5 \quad \mathcal{O}(10^2)\text{--}\mathcal{O}(10^4)$$

Remark 5.5. REX ”” SCX T T

Part V

MCMC

Part V: MCMC Convergence Diagnostics

6

Formalizing Audit Convergence

6.1

SCX ””

-
-
-

SCX g_i $i \sum_j g_{ij}$ g_{ij} Gelman-Rubin $\hat{R} < 1.01$

6.2 Gelman-Rubin

Definition 6.1 (Gelman-Rubin $\hat{R}$). $M \geq 2$ N θ
(within-chain variance):

$$W = \frac{1}{M} \sum_{m=1}^M s_m^2, \quad s_m^2 = \frac{1}{N-1} \sum_{n=1}^N (\theta_{mn} - \bar{\theta}_m)^2 \quad (27)$$

(between-chain variance):

$$B = \frac{N}{M-1} \sum_{m=1}^M (\bar{\theta}_m - \bar{\theta})^2, \quad \bar{\theta} = \frac{1}{M} \sum_{m=1}^M \bar{\theta}_m \quad (28)$$

:

$$\widehat{\text{Var}}^+(\theta) = \frac{N-1}{N} W + \frac{1}{N} B \quad (29)$$

$\hat{R}$:

$$\hat{R} = \sqrt{\frac{\widehat{\text{Var}}^+(\theta)}{W}} \quad (30)$$

Theorem 6.2 (Gelman-Rubin). $M \widehat{\text{Var}}^+ W \hat{R} \approx 1$

$$\iff \max_i \hat{R}(g_i) < 1.01 \quad (31)$$

$\max_i$

Remark 6.3. $\hat{R} < 1.01$ $\hat{R} < 1.1$ SCX (a) (b) $g_i \sum g = 0$

6.3

Definition 6.4 (Effective Sample Size). $\rho_k = (\theta_t, \theta_{t+k})$

$$\text{ESS} = \frac{N}{1 + 2 \sum_{k=1}^{\infty} \rho_k} \quad (32)$$

$$\text{ESS} \approx N / (1 + 2 \sum_{k=1}^{K_{\max}} \hat{\rho}_k) \quad K_{\max} \quad \hat{\rho}_k \quad k$$

Proposition 6.5 (SCX ESS). *SCX*

$$(1) \quad \hat{g}_{ij} \quad \text{ESS} \geq 100$$

$$(2) \quad \widehat{\text{Var}}[g_{ij}] \text{ESS} \geq 400$$

$$(3) \quad \text{ESS} \geq 1000$$

$$\mathcal{O}(\text{ESS}^{-1/2}) \quad \mathcal{O}(\text{ESS}^{-1/2})$$

6.4

SCX Multi-Chain Initialization for SCX Audit

$\hat{\mathbf{R}}$

1. **1** $g_{ij} \sim \mathcal{N}(0, 0.1)$ ——
2. **2** $g_{ij} \sim \mathcal{N}(0, 1.0)$ ——
3. **3** $g_{ij} \sim \text{SkewNormal}(0, 0.5, \alpha = 3)$ ——
4. **4** $g_{ij} \sim \text{SkewNormal}(0, 0.5, \alpha = -3)$ ——
5. **5-8** $g_{ij} \sim \mathcal{N}(0, \sigma_j^2) \quad \sigma_j \sim \text{Exp}(1)$ ——

$M \quad \hat{\mathbf{R}}$

6.5 Rank

$\hat{\mathbf{R}} \quad \text{ESSRank}$

Definition 6.6 (Rank Rank-Normalized Diagnostic). $M \quad \theta \quad \text{Rank} \quad r_{mn} = \text{rank}(\theta_{mn}) \quad 1$
 $MN \quad [0, 1] z_{mn} = (r_{mn} - 0.5) / (MN)$
 $z_{mn} \quad [0, 1] \quad \text{Kolmogorov-Smirnov}$

Theorem 6.7 (Composite Diagnostic). *SCX fully converged*

$$\left\{ \begin{array}{ll} \max_i \hat{\mathbf{R}}(g_i) < 1.01 & (\text{Gelman-Rubin}) \\ \min_i \text{ESS}(g_i) > 400 & () \\ \max_i \text{KS-}p\text{-value} > 0.05 & (\text{Rank}) \\ & () \end{array} \right. \quad (33)$$

6.6

Algorithm 6.8 (Adaptive Early Stopping). 1: $\theta \leftarrow N_{\min} w$

```

2:  $n \leftarrow 0$   $\hat{R} \leftarrow []$ 
3: while  $n < N_{\min}$  do
4:    $M \leftarrow L$ 
5:    $n \leftarrow n + L$ 
6:    $w \leftarrow \hat{R}(\theta)$ 
7:    $\mathcal{R}.\text{append}(\hat{R}(\theta))$ 
8:   if  $|\mathcal{R}| \geq 3$  then
9:      $\hat{R}_t = a \cdot t^{-b} + 1$ 
10:     $a < 0.01$   $b > 0$   $\hat{R} \leftarrow 1$  break
11:   end if
12: end while
13:  $\hat{R} \leftarrow \text{ESS}$ 

```

6.7 SCX

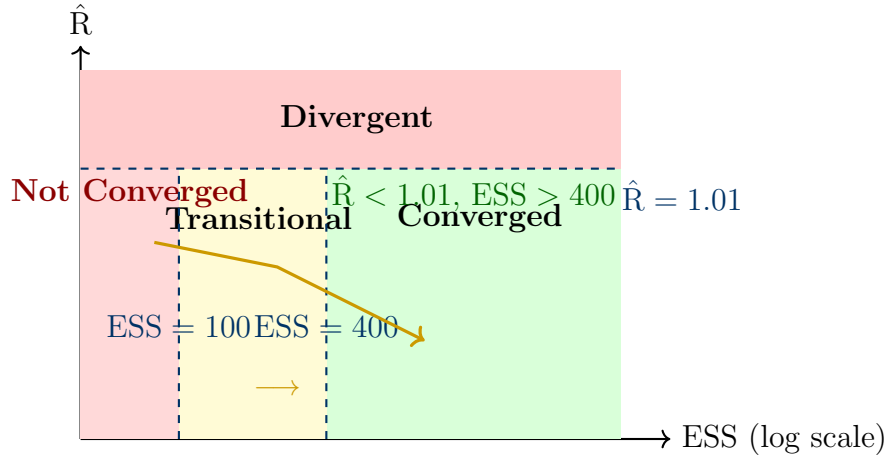


Figure 1: SCX $\hat{R}$ vs ESS

7

Method Integration and Audit Pipeline

7.1

SCX

- 1: **(TI):** Cercis $\mathfrak{C}_{\text{cal}}$
- 2: **(REX):**
- 3: **(HMC):** Situs HMC β_{target} Boltzmann
- 4: **(SMC):** HMC SMC
- 5: **($\hat{R}$ /ESS):** $\hat{R}$ ESS Rank

7.2

Table 1:			
/			
Situs-HMC	$\mathcal{O}(mn)$	$M \sim 4-8$	$\mathcal{O}(Mmn \cdot N)$
SCX-SMC	$\mathcal{O}(Kmn)$	$K \sim 10^3-10^4$	$\mathcal{O}(Kmn \cdot T)$
TI	$\mathcal{O}(Rmn \cdot N)$	$R \sim 10-50$	$\mathcal{O}(Rmn \cdot N)$
REX	$\mathcal{O}(RLmn)$	$R \sim 5-20$	$\mathcal{O}(RLmn \cdot N_{\text{iter}})$
	$\mathcal{O}(N)$	—	$\mathcal{O}(N \log N)$

$m = n = K = R = L = N = T = \text{SMC}$

Remark 7.1. Cercis $\mathcal{O}(mn)$ Situs $\mathcal{O}(m^2n^2)$ $m \sim 10^4$, $n \sim 10^3$ Situs

7.3

$m = 5$ $n = 3$ verify_monte_carlo.py

- (1) **HMC:** Situs-HMC ESS HMC 2–5
- (2) **SMC:** $t \rightarrow T$ ESS $K/2$ $\alpha = 0.5$
- (3) **TI:** $\mathfrak{C}_{\text{cal}}$ Monte Carlo $< 5\%$
- (4) **REX:** $\mathfrak{C}$
- (5) $\hat{R}$: g_i $\hat{R} < 1.01$

A `verify_monte_carlo.py`

Verification Script: `verify_monte_carlo.py`

Python numpy SCX

`verify_monte_carlo.py`

(Refer to accompanying file: `verify_monte_carlo.py`)

A.1

1: **Situs** $\otimes$ Situs Cercis

2: **HMC** HMC + Situs-HMC

3: **SMC** ESS

4: **TI** + +

5: **REX** +

6: $\hat{\text{RESSRank}}$

7: +

B

Key Formulas Reference

$\pi_\beta(E) \propto e^{-\beta \mathfrak{E}(E)}$	Boltzmann
$H(q, p) = \frac{1}{2} p^\top M^{-1} p + \beta \mathfrak{E}(q)$	Situs-HMC
$\dot{p} = -\nabla U + \frac{1}{2} \nabla(p^\top M^{-1} p)$	
$\text{ESS}_t = (\sum w_k^2)^{-1}$	SMC
$\Delta F = \int \mathbb{E}_{\pi_\beta}[\mathfrak{E}] d\beta$	
$A_{\text{swap}} = \min(1, e^{\Delta\beta \cdot \Delta\mathfrak{E}})$	REX
$\hat{\text{R}} = \sqrt{\widehat{\text{Var}}^+} / W$	Gelman-Rubin
$\text{ESS} = N / (1 + 2 \sum \rho_k)$	

Acknowledgments

SCX

Duane et al. (1987) HMC Gordon et al. (1993) Gelman & Rubin (1992) Swendsen & Wang (1986)

We acknowledge the foundational contributions of Duane et al. (1987) for HMC, Gordon et al. (1993) for particle filters, Gelman & Rubin (1992) for convergence diagnostics, and Swendsen & Wang (1986) for replica exchange methods. The novelty of this work lies not in inventing these methods, but in their systematic application to SCX audit problems — and in demonstrating that the Situs manifold provides the natural geometric setting for this application.

All things are samplable. All phenomena converge.

— SCX Monte Carlo, 2026-07-02